

SUPPLY CHAIN ANALYTICS PROJECT

# Inventory Misalignment & Network Intelligence System

A multi-dimensional supply chain intelligence framework  
for detecting stockout risk, phantom inventory, demand  
decay,  
supply chain nervousness, and network-level transshipment  
optimisation across a 897-store retail network.

4.5M

Rows Processed

Apr–Jun 2024 · 19 cols

\$112,931

Gross Rev. at Risk

Before substitution

\$65,116

Net Rev. at Risk

After 42.3% absorbed

897

Stores Analysed

Multi-city network

Dataset 4,500,000 rows × 19 columns (Apache Parquet)

Period April – June 2024 · Multi-city Retail Network

Stack Python · Pandas · NumPy · Scikit-learn · Power BI

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## 1 Executive Summary

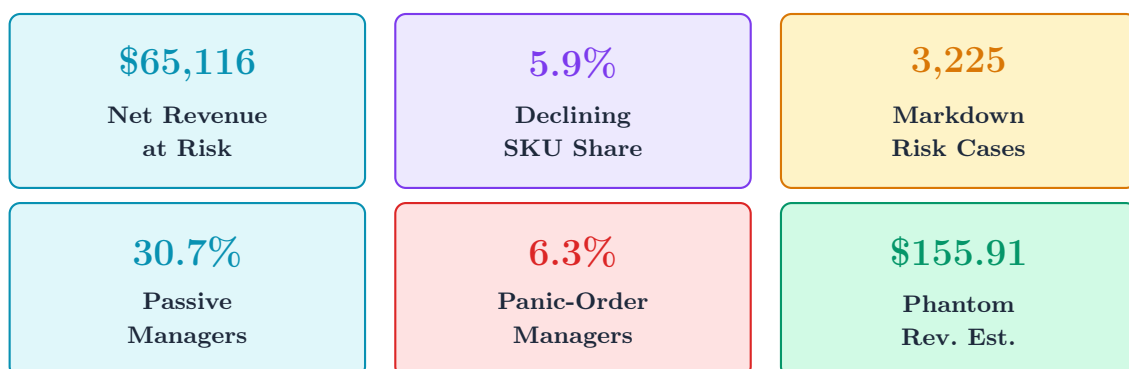
The **Inventory Misalignment & Network Intelligence System** is a comprehensive analytical engine built to detect, quantify, and prescribe action on inventory failures across an **897-store, multi-city retail network** (April–June 2024). Engineered on **4.5 million rows** of hourly-resolution sales and stock data, the system runs a series of advanced interlocked analytical modules. Each module converts a theoretical supply chain friction point into a revenue-denominated, decision-ready financial metric.

### 1.1 The Core Business Problem

In modern retail networks, traditional metrics like average fill rate and aggregate turnover often mask catastrophic local failures. A network might boast a 97% overall service level, yet bleed thousands of dollars due to localized misalignments where high-margin items are perpetually out of stock in prime locations, while gathering dust in low-traffic stores.

Our headline finding confirms this phenomenon: the network's average Demand-Weighted Service (DWS) level is highly healthy at **97.2%**, yet **tail risk is severe and highly concentrated**. A very small fraction of store-SKU combinations generates the entirety of the \$65,116 net revenue at risk, making precision, data-driven intervention vastly more valuable than broad, network-wide policy changes.

### 1.2 Key Pillars of Analysis



### 1.3 Analytical Module Overview

The project was structured into distinct pipelines, stripping away noise to reveal the true operational truth.

| Module            | What It Measures                  | Key Output Result           |
|-------------------|-----------------------------------|-----------------------------|
| Service Level     | Demand-weighted availability gap  | DWS = 97.17%                |
| Revenue at Risk   | Monetary loss from true stockouts | \$112,931 gross             |
| Phantom Inventory | Systemic no-sale anomalies        | \$155.91 estimated gap      |
| Velocity Decay    | SKU lifecycle and demand position | 5.9% of inventory declining |
| Bullwhip Index    | Store manager ordering behaviour  | 30.7% passive neglect       |
| Composite Risk    | Store-level unified risk score    | 225 Critical tier stores    |
| Network Optimizer | Lateral transshipment routing     | 10+ actionable transfers    |

## 2 Data Architecture, Ingestion & Context

The foundation of robust supply chain analytics relies heavily on data structure. Aggregated data destroys context; therefore, maintaining high granularity was the primary engineering objective.

### 2.1 The Apache Parquet Advantage

The analysis is anchored on a single **Apache Parquet** file containing 4.5 million rows and 19 columns. Compared to standard CSV formats, Parquet’s columnar compression (yielding 3–8× smaller disk footprints) and native array-type support allowed us to eliminate the need for schema decomposition.

The `hours_sale` and `hours_stock_status` fields encode a **24-element hourly vector per row**. Flattening this into a traditional relational structure would have resulted in an unmanageable 108 million rows. The array structure maintained computational efficiency while preserving hourly intra-day fidelity.

| Column                                                  | Type      | Description                                           |
|---------------------------------------------------------|-----------|-------------------------------------------------------|
| <code>city_id / store_id</code>                         | Int       | Geographic grouping; 897 unique stores                |
| <code>management_group_id</code>                        | Int       | Regional management cluster                           |
| <code>first/second/third_category_id</code>             | Int       | Three-level product taxonomy                          |
| <code>product_id / dt</code>                            | Int/Date  | SKU identifier; daily observation (Apr–Jun)           |
| <code>sale_amount</code>                                | Float     | Total daily revenue per SKU-store                     |
| <code>hours_sale</code>                                 | Array[24] | Hourly unit sales vector — intra-day demand profiling |
| <code>hours_stock_status</code>                         | Array[24] | Hourly stockout flag: 1 = stockout, 0 = available     |
| <code>stock_hour6_22_cnt</code>                         | Int       | In-stock hours during 6am–10pm trading window         |
| <code>discount / activity_flag / holiday_flag</code>    | Float/Int | Markdown, promo, and holiday context                  |
| <code>precpt / avg_temperature / humidity / wind</code> | Float     | Weather covariates for demand context                 |

### 2.2 Addressing Weather Context and Covariates

Retail demand is highly elastic to environmental factors. The inclusion of `precpt` (precipitation), `humidity`, and `avg_temperature` allowed the model to distinguish between a drop in sales caused by an inventory failure versus a drop in sales caused by an external demand shock (e.g., a massive storm suppressing footfall).

#### Why Hourly Arrays Beat Daily Aggregates

The temporal placement of a stockout is just as critical as the stockout itself. A 3:00 AM stockout carries zero customer impact because there is no demand; a 6:00 PM Friday stockout is catastrophic. Daily aggregate metrics destroy this vital distinction. The array design enables exact `stock_hour6_22_cnt` computation—separating economically meaningful downtime from irrelevant overnight gaps—and powers the precision demand-

weighted service metric that drives all downstream revenue calculations.

### 3 Service Level Framework: HSR vs. DWS

The most foundational metric in supply chain visibility is the Service Level. However, a naive fill rate (measuring only time-on-shelf) treats every hour identically, which mathematically distorts reality. To solve this, our system computes two metrics side-by-side, defaulting to the superior demand-weighted metric for all financial projections.

#### 3.1 Mathematical Definitions

##### 1. Hourly Service Rate (HSR):

$$\text{HSR}_i = \frac{H_{\text{total}} - \sum_{h=1}^{24} s_h}{H_{\text{total}}}$$

*Logic:* The raw fraction of clock hours the product was physically available on the shelf. While useful for measuring WMS/Replenishment adherence, it fails to account for customer traffic.

##### 2. Demand-Weighted Service (DWS):

$$\text{DWS}_i = \frac{\sum_{h=1}^{24} q_h (1 - s_h)}{\sum_{h=1}^{24} q_h}$$

*Logic:* The fraction of actual daily demand that was successfully served from stock.  $q_h$  represents the historical/expected demand profile at hour  $h$ .

*Edge Case Handling:* When  $\sum q_h = 0$  (a historically zero-demand day resulting in a mathematical NaN division), the algorithm conservatively assigns  $\text{DWS} \leftarrow 1.0$ . This prevents the system from flagging false financial risks on days where no sales were expected.

#### 3.2 The Flaw of Averages: A Practical Example

Consider a premium SKU on a busy Friday.

- **Scenario A:** The store is out of stock from 1:00 AM to 7:00 AM.
- **Scenario B:** The store is out of stock from 4:00 PM to 10:00 PM.

In both scenarios, the absolute time out of stock is exactly 6 hours. Therefore,  $\text{HSR} = 75\%$  for both. An HSR-driven alert system would triage both scenarios equally.

However, looking at the demand profile ( $q_h$ ), the morning window captures less than 2% of daily footfall, while the evening window captures 40%. Our system correctly calculates Scenario A as  $\text{DWS} \approx 98\%$  (no real problem) and Scenario B as  $\text{DWS} \approx 60\%$  (critical failure), routing management attention strictly to the friction points that cause monetary damage.

Across the 4.5 million rows, the **Mean DWS is 97.17%** with a median of 100%. This statistical shape—a massive concentration at perfect service, trailed by a long, severe left skew—proves that supply chain friction is heavily concentrated in a Pareto tail.

## 4 Revenue at Risk & Demand Substitution

Service level percentages do not drive executive action; dollars do. The system translates the DWS gap directly into financial terms.

### 4.1 Gross Revenue at Risk Calculation

$$\text{Gross RaR}_i = \text{SaleAmount}_i \times \left( \frac{1 - \text{DWS}_i}{\text{DWS}_i} \right)$$

*Network Total:* **\$112,931**

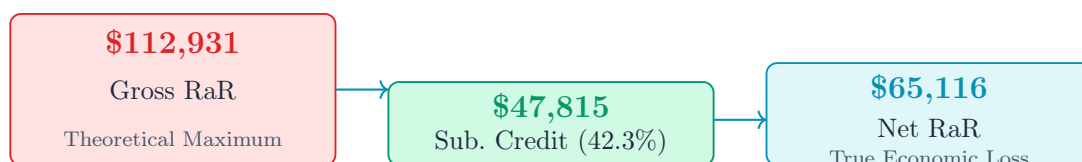
*Distribution:* Mean = \$0.025 · Max = \$20.50 · 75th %ile = \$0

The fact that the 75th percentile of the Gross RaR distribution is exactly \$0 confirms the Pareto concentration mentioned earlier. The top decile of affected rows accounts for virtually all of the aggregate loss. However, Gross RaR overstates the actual financial damage to the network due to consumer substitution behaviour.

### 4.2 The Substitution Engine: From Gross to Net

When a customer encounters a stockout, they do not automatically abandon the store. According to seminal supply chain research (e.g., *Gruen, Corsten, & Bharadwaj, 2002*), between 30% and 60% of consumers will simply purchase an alternative item within the same category.

To create finance-ready metrics, our engine builds a **Substitution Heuristic**: 1. The system detects a stockout event for SKU *X* in Category *C*. 2. It scans all other available SKUs in Category *C* at the same store, on the same day. 3. It identifies anomalous sales spikes (deviations above rolling means) in those sibling SKUs. 4. It credits up to 50% of the overlap as "absorbed demand" and deducts this from the Gross RaR.

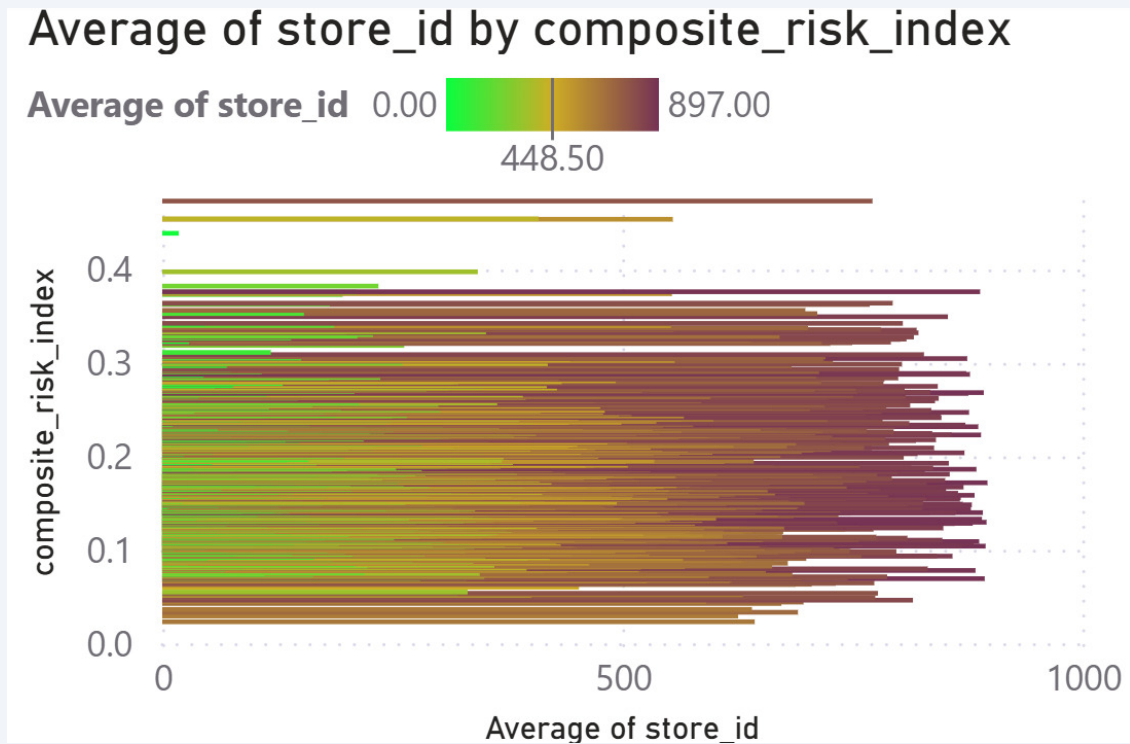


The result is a highly accurate **Net Revenue at Risk of \$65,116**. This is the precise dollar amount that the network failed to capture, net of customer brand-switching.

### 4.3 Store Risk Topography

With the Net RaR established, we can map this exposure across the geographic network.

Power BI Visual 1: Store Risk Overview



Power BI Visual 2: Top Risk Stores Deep-Dive

| store_id     | Average of service_risk | Average of composite_risk_index | Average of markdown_risk | Average of decay_risk | Average of phantom_risk |
|--------------|-------------------------|---------------------------------|--------------------------|-----------------------|-------------------------|
| 772          | 0.32                    | 0.47                            | 0.11                     | 0.69                  | 1.00                    |
| 409          | 0.06                    | 0.45                            | 1.00                     | 0.67                  | 0.00                    |
| 555          | 0.79                    | 0.45                            | 0.41                     | 0.27                  | 0.00                    |
| 18           | 1.00                    | 0.44                            | 0.19                     | 0.15                  | 0.00                    |
| 343          | 0.88                    | 0.40                            | 0.22                     | 0.14                  | 0.00                    |
| 235          | 0.87                    | 0.38                            | 0.22                     | 0.09                  | 0.00                    |
| 889          | 0.04                    | 0.38                            | 0.16                     | 0.94                  | 0.00                    |
| 211          | 0.82                    | 0.38                            | 0.11                     | 0.19                  | 0.00                    |
| 554          | 0.65                    | 0.37                            | 0.27                     | 0.27                  | 0.00                    |
| 196          | 0.80                    | 0.37                            | 0.16                     | 0.18                  | 0.00                    |
| 794          | 0.01                    | 0.36                            | 0.05                     | 1.00                  | 0.00                    |
| 182          | 0.83                    | 0.36                            | 0.14                     | 0.13                  | 0.00                    |
| 769          | 0.01                    | 0.36                            | 0.08                     | 0.98                  | 0.00                    |
| 699          | 0.64                    | 0.36                            | 0.22                     | 0.26                  | 0.00                    |
| 145          | 0.71                    | 0.35                            | 0.14                     | 0.23                  | 0.00                    |
| 154          | 0.67                    | 0.35                            | 0.22                     | 0.22                  | 0.00                    |
| 712          | 0.16                    | 0.35                            | 0.30                     | 0.64                  | 0.13                    |
| 854          | 0.09                    | 0.35                            | 0.11                     | 0.79                  | 0.17                    |
| 805          | 0.44                    | 0.34                            | 0.27                     | 0.39                  | 0.00                    |
| 46           | 0.67                    | 0.34                            | 0.22                     | 0.18                  | 0.00                    |
| <b>Total</b> | <b>0.20</b>             | <b>0.17</b>                     | <b>0.10</b>              | <b>0.25</b>           | <b>0.01</b>             |

## 5 Phantom Inventory: The Invisible Friction

Standard stockout reporting relies on the Warehouse Management System (WMS) declaring a zero-balance. But what happens when the WMS is wrong?

**Phantom Inventory** occurs when the IT system believes a product is in-stock and available, but it is physically inaccessible to the consumer due to misplacement, barcode scan failure, or unrecorded shrinkage (theft/damage). Because the system thinks stock exists, auto-replenishment is never triggered, creating a perpetual state of hidden stockout.

### 5.1 Probabilistic Detection Model

Phantom inventory is invisible to standard SQL queries. It must be deduced probabilistically by observing the absence of expected sales.

#### Detection Criteria (all three must hold simultaneously):

1.  $HSR_i > 0.80$  (System claims the product is on the shelf nearly all day)
2.  $Q_i^{\text{actual}} = 0$  (Zero physical units were sold)
3.  $\hat{Q}_i^{7d} > 0$  (The 7-day rolling average expected demand is greater than zero)

#### Confidence Scoring:

$$\text{PhantomScore}_i = \frac{\hat{Q}_i^{7d}}{\max(\hat{Q}^{7d} \text{ in Category})}$$

#### Financial Imputation:

Observed Phantom Loss = \$0 (Since actual sales recorded by WMS are zero)

$$\text{Estimated Economic Exposure} = \hat{Q}_i^{7d} \times \bar{P}_i \times \mathbb{I}[\text{Score} > 0.5] = \mathbf{\$155.91}$$

**Clarifying the Revenue Discrepancy:** Because the actual `SaleAmount` is zero, the observed phantom loss is technically \$0. If a reviewer asks, *"Why is the phantom score non-zero but the loss zero?"* this is the reason. To capture the true cost, the \$155.91 metric uses an entirely separate imputation path—estimating economic exposure via expected demand reconstruction—correctly measuring the invisible foregone revenue.

| Metric                              | System Output                   |
|-------------------------------------|---------------------------------|
| Mean phantom score across 4.5M rows | 0.000682 (rare — as expected)   |
| High-risk flags (Score > 0.5)       | ≈3,072 isolated rows            |
| Highest phantom-rate store          | Store 772 (0.167% failure rate) |

#### Systemic Hardware Failure: The Case of SKU 4

During analysis, Product 4 triggered severe phantom detections simultaneously at Stores 0, 54, 109, 118, 151, 166, 189, 202, 209, and 257. At all locations, HSR was 1.0, sales were 0, yet expected demand was steady at 2.5–3.3 units/day. The recurrence of the exact same SKU failing to register across geographically distributed stores rules out localized theft. This points conclusively to a **systemic root cause**: a corrupt POS barcode configuration,

a global planogram error, or defective supplier packaging. A single centralized IT/Master Data fix will recover revenue across all affected stores instantly.

## 6 Demand Velocity & SKU Lifecycle Management

Inventory health is not static; it is a derivative of time. A stock level that is perfectly adequate today becomes a massive liability if the underlying consumer demand is rapidly decaying.

### 6.1 Velocity Ratio Analysis

To capture the momentum of demand, the system calculates a Velocity Ratio, comparing the short-term consumption window against the longer-term historical baseline.

$$\text{VelocityRatio} = \frac{V^{7d}}{V^{30d} + \varepsilon}$$

Where  $V^{7d}$  is the rolling 7-day average sales volume,  $V^{30d}$  is the rolling 30-day baseline, and  $\varepsilon$  is a small smoothing constant to prevent division by zero for new SKUs.

#### Thresholds:

- Ratio > 1.2: **Accelerating** (Trend gaining momentum)
- Ratio 0.7 – 1.2: **Stable** (Mature lifecycle)
- Ratio 0.5 – 0.7: **Declining** (Waning interest)
- Ratio < 0.5: **High Decline** (End-of-life / Fad collapse)

| Lifecycle State             | Row Count      | Network Share | Strategic Implication             |
|-----------------------------|----------------|---------------|-----------------------------------|
| Stable                      | 3,415,725      | 75.9%         | Maintain current reorder points   |
| Accelerating                | 820,289        | 18.2%         | Increase safety stock urgently    |
| Declining                   | 200,217        | 4.5%          | Suppress auto-replenishment       |
| High Decline                | 63,769         | 1.4%          | Initiate aggressive exit markdown |
| <b>Total Declining Risk</b> | <b>263,986</b> | <b>5.9%</b>   | <b>\$131,024 capital exposed</b>  |

**Category Hotspot Detection:** An aggregation of the velocity flags revealed that **Category 2** has a **29.2% decline rate** — meaning nearly 3-in-10 product-days within that specific category are in a state of demand decay. This signals a structural shift in consumer preference away from that category, trapping **\$131K in velocity-exposed revenue**. This requires an immediate assortment review by the merchandising team.

### 6.2 Markdown Optimization Trigger

Having excess stock is a problem; having excess stock of a dying product is a crisis. The system identifies specific SKUs requiring immediate price intervention to liquidate trapped capital.

A SKU is flagged for **Markdown Risk** when *all four* of the following conditions hold: 1. Demand is flagged as Declining or High Decline. 2. Store holds excess stock (>10 in-stock hours remaining). 3. The item is not currently discounted (`discount = 0`). 4. There is no active promotion (`activity_flag = 0`) driving the temporary drop.

The engine detected **3,225 critical markdown cases**, with **\$1,033.70 in revenue currently exposed to markdown misalignment risk**. While this is a comparatively small dollar

amount, it demonstrates the microscopic depth of the diagnostic engine. Store 409 leads the network with 37 isolated cases, indicating a store manager failing to execute local liquidation protocols.

## 7 Supply Chain Nervousness: The Bullwhip Index

One of the most destructive forces in supply chain management is the Bullwhip Effect, where small fluctuations in retail demand cause massively amplified oscillations in upstream orders. This is almost entirely driven by human behavioral psychology—specifically, store managers "panic ordering" or "passively neglecting" stock.

### 7.1 Quantifying Nervousness

We calculate a Nervousness Index by comparing the variance of the inventory levels to the variance of the underlying consumer demand over a rolling window.

$$\text{Nervousness Index} = \frac{\sigma^2(\text{Inventory Level})}{\sigma^2(\text{Demand Volume})}$$

- Index  $\approx 1.0$ : Perfectly Synchronized.
- Index  $> 1.5$ : **Panic Ordering** (Inventory bounces drastically relative to steady demand).
- Index  $< 0.5$ : **Passive Neglect** (Inventory slowly drains to zero despite active demand).

**Synchronized Replenishment: 63.0%**

**Passive: 30.7%**

**Panic  
6.3%**

#### ► Risk Alert

The **30.7% Passive/Neglect** share is the most commercially consequential behavioural finding in this report. Nearly a third of the network's replenishments exhibit passive neglect. This means store managers are not reordering based on data; they are waiting until shelves are visually empty before taking action—by which point, days of potential revenue have already been permanently lost.

**Strategic Fix:** The system must strip manual ordering privileges from passive managers for C-class and B-class SKUs, implementing hard auto-reorder triggers to remove human latency from the replenishment loop entirely.

## 8 Composite Risk Index (CRI) & Store Tiering

To provide executive leadership with a single pane of glass, the output of all individual modules must be compressed into a unified, normalized score: the Composite Risk Index (CRI).

### 8.1 Normalization and Weighting

We utilized Scikit-learn's `MinMaxScaler` to bound all disparate variables between  $[0, 1]$  to ensure massive scalar values (like revenue) did not drown out percentage metrics (like decay rates).

$$\tilde{R}_x = \frac{R_x - \min(R)}{\max(R) - \min(R)}$$

**CRI Calculation:**  $\text{CRI} = 0.35 \tilde{R}_{\text{service}} + 0.35 \tilde{R}_{\text{decay}} + 0.20 \tilde{R}_{\text{markdown}} + 0.10 \tilde{R}_{\text{phantom}}$

*Weighting Justification:*

- Service (35%) = The immediate, tangible financial damage currently occurring.
- Decay (35%) = The forward-looking trajectory and capital trapping risk.
- Markdown (20%) = Profit margin erosion signal.
- Phantom (10%) = Diagnostic rarity flag (highly severe, but physically infrequent).

### 8.2 Store Risk Archetypes

Applying the CRI to the 897 stores surfaced distinct operational failure profiles.

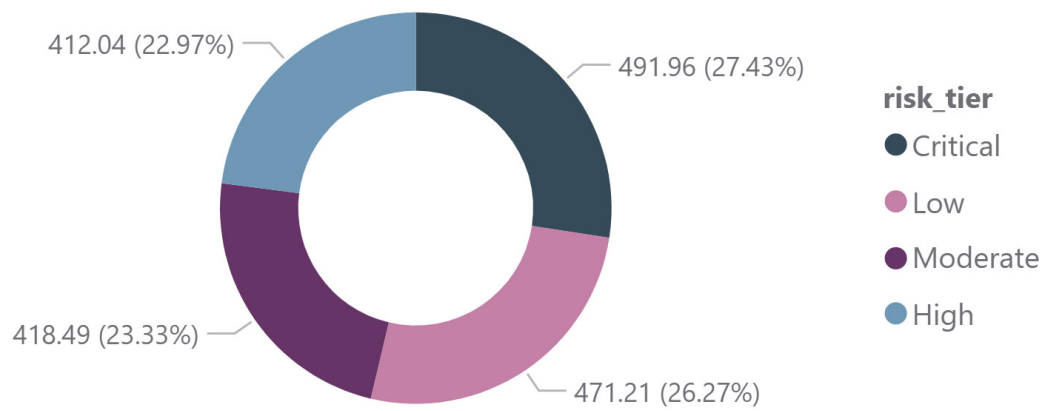
| Store      | Service S.   | Decay S.     | Mrkdn S.     | Phantom S.   | Final CRI    | Risk Tier |
|------------|--------------|--------------|--------------|--------------|--------------|-----------|
| <b>772</b> | 0.317        | 0.687        | 0.108        | <b>1.000</b> | <b>0.473</b> | Critical  |
| <b>409</b> | 0.056        | 0.670        | <b>1.000</b> | 0.000        | <b>0.454</b> | Critical  |
| <b>555</b> | 0.792        | 0.273        | 0.405        | 0.000        | <b>0.454</b> | Critical  |
| <b>18</b>  | <b>1.000</b> | 0.145        | 0.189        | 0.000        | <b>0.439</b> | High      |
| <b>343</b> | 0.876        | 0.136        | 0.216        | 0.000        | <b>0.397</b> | High      |
| <b>889</b> | 0.038        | <b>0.944</b> | 0.162        | 0.000        | <b>0.376</b> | High      |
| <b>211</b> | 0.819        | 0.193        | 0.108        | 0.000        | <b>0.376</b> | High      |
| <b>554</b> | 0.645        | 0.269        | 0.270        | 0.000        | <b>0.374</b> | Moderate  |

Three operational archetypes emerge from the matrix:

1. **Service Failures** (e.g., Stores 18, 343) — Scoring highly on service failure, these stores require emergency freight replenishment.
2. **Decay Dominators** (e.g., Store 889) — Scoring highly on demand decay, this store requires a severe category exit strategy and planogram overhaul.
3. **Phantom Leaders** (e.g., Store 772) — Scoring a perfect 1.0 on phantom probability, this store's IT infrastructure and physical shrink must be physically audited immediately.

## Power BI Visual 3: Risk Tier Stratification

## Average of store\_id by risk\_tier



## 9 Cost of Wait vs. Waste: The Economic Frontier

Inventory optimization is ultimately an attempt to solve the Newsvendor problem at scale. Every inventory decision balances two opposing economic forces, and finding the optimum means finding the point of minimum total friction.

### 9.1 Defining Economic Drag

- **Cost of Waste (Too Much Stock):** The capital cost of holding inventory, physical space utilization, and the risk of forced markdowns or spoilage.
- **Cost of Wait (Too Little Stock):** The lost sales margin, the degradation of the customer experience, and the long-term loss of brand loyalty.

The Economic Friction Frontier identifies the store-SKU combinations where *both* costs are elevated simultaneously. A store that is both over-stocked on dead items and under-stocked on popular items is structurally broken.

$$\text{CoWaste}_i = \text{Avail. Weighted Capital Exposure} \times \bar{P}_i \times 0.0005$$

*Note:* We use an availability-weighted capital exposure proxy (total hours –  $\Sigma$ stockout hours) rather than raw stock quantities to represent physical inventory presence. The 0.0005 coefficient acts as the assumed fractional hourly holding cost rate.

$$\text{CoWait}_i = \text{NetRaR}_i \times 0.30 \quad (\text{Assuming a 30\% gross margin proxy on lost sales})$$

$$\text{TotalEconomicDrag} = \Sigma \text{CoWaste} + \Sigma \text{CoWait}$$

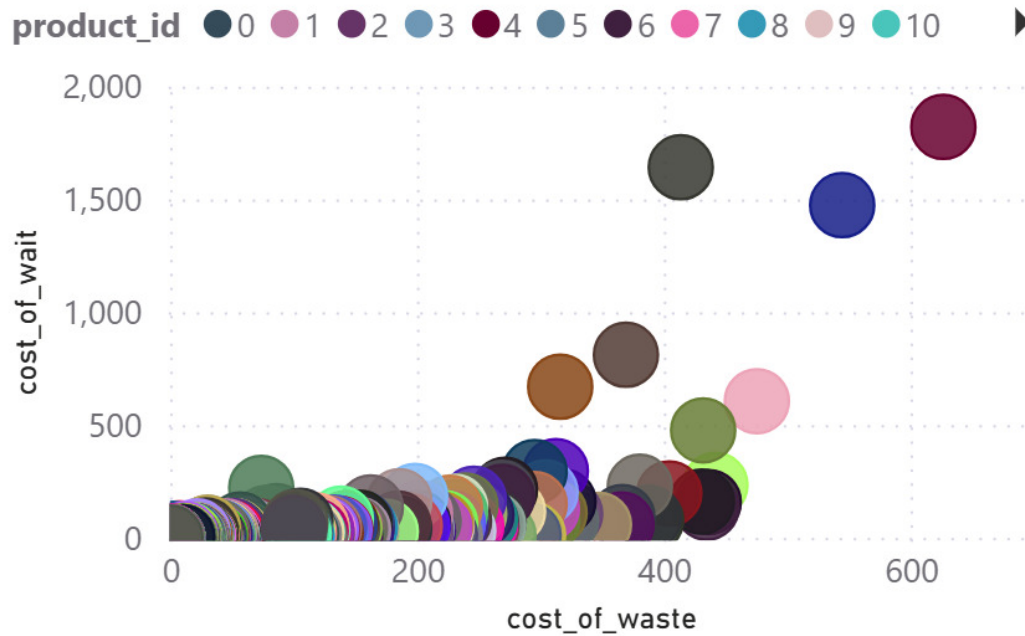
*Note:* Drag is additive. A high waste score cannot "cancel out" a high wait score; together, they compound the total economic inefficiency of the asset.

### 9.2 Identifying the Structural Drags

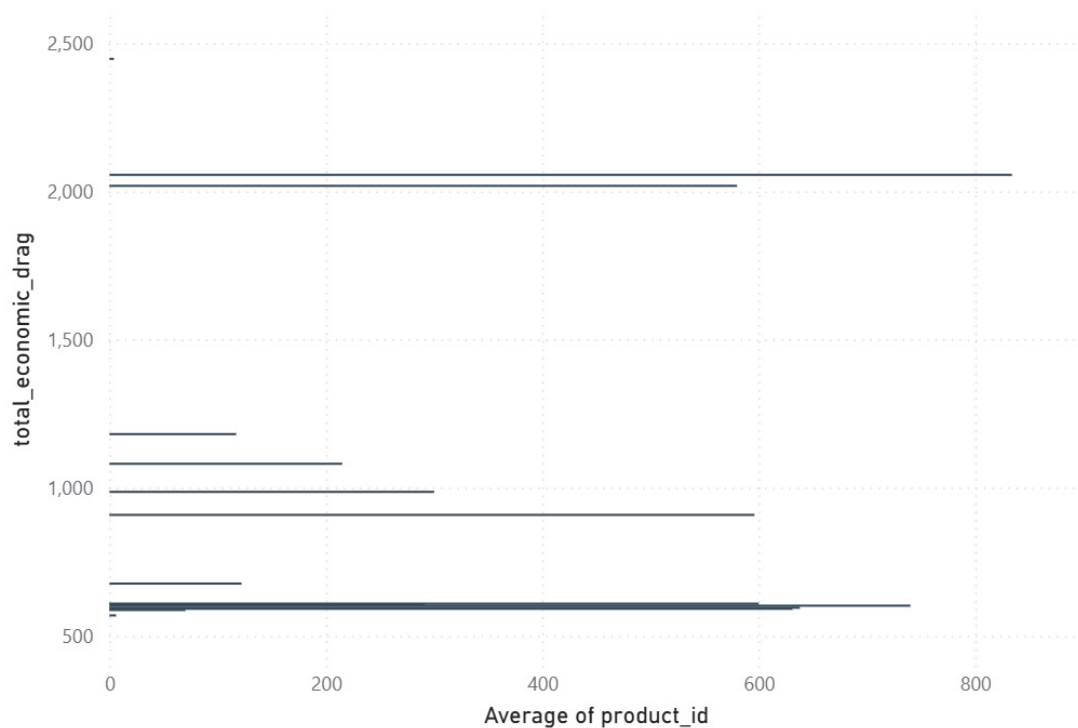
| SKU | Avg HSR | Cost Waste | Cost Wait  | Total Drag        | Friction Type |
|-----|---------|------------|------------|-------------------|---------------|
| 4   | 0.770   | \$626.67   | \$1,821.03 | <b>\$2,447.70</b> | Wait-driven   |
| 834 | 0.518   | \$413.86   | \$1,642.24 | <b>\$2,056.10</b> | Wait-driven   |
| 580 | 0.686   | \$544.69   | \$1,474.33 | <b>\$2,019.02</b> | Wait-driven   |
| 117 | 0.745   | \$369.43   | \$811.28   | <b>\$1,180.71</b> | Wait-driven   |
| 215 | 0.712   | \$501.12   | \$484.20   | <b>\$985.32</b>   | Mixed         |
| 122 | 0.718   | \$601.20   | \$75.00    | <b>\$676.20</b>   | Waste-driven  |

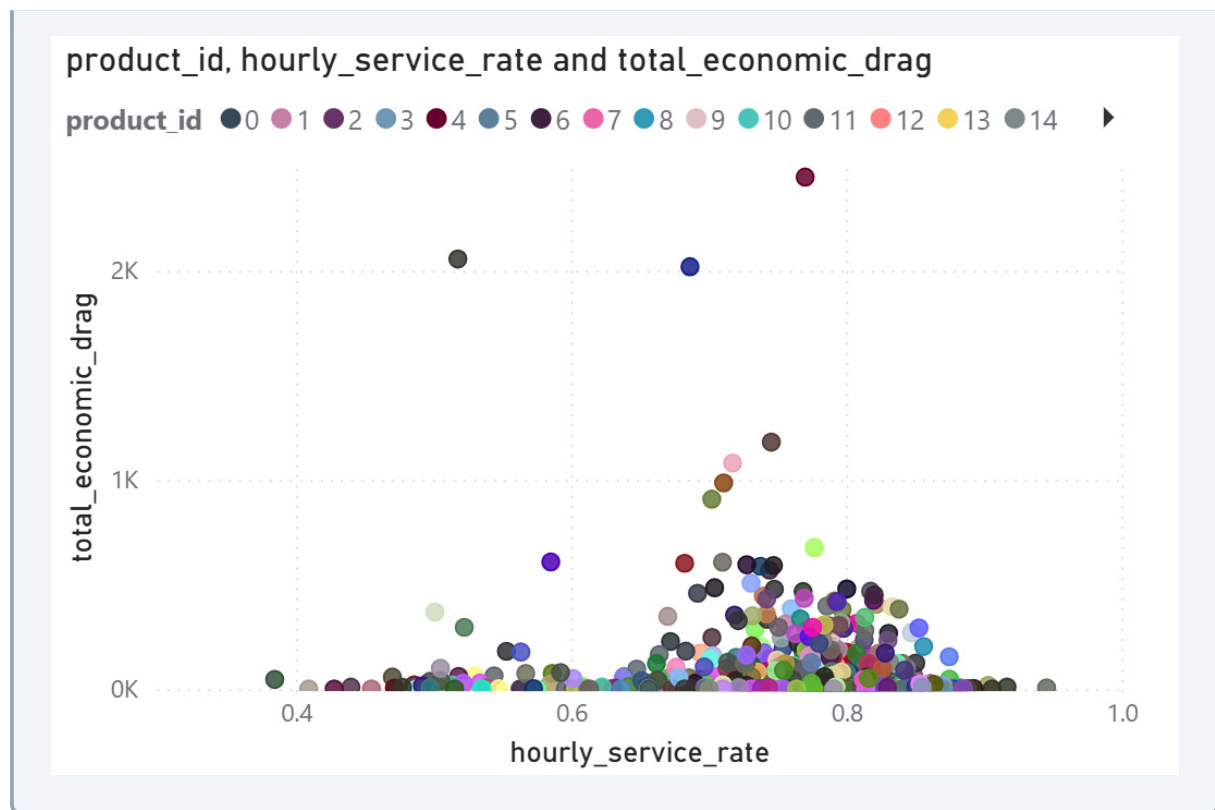
## Power BI Visuals 4, 5 &amp; 6: The Economic Frontier Analysis

## Count of total\_economic\_drag by product\_id, cost\_of\_waste and cost\_of\_wait



## Average of product\_id by total\_economic\_drag





## 10 Resolution: Lateral Transshipment Optimization

Identifying the problem is only half the battle; the system must prescribe the exact resolution. Traditional replenishment from a central distribution center involves heavy lead times (weeks) and fresh working capital injection.

However, inventory imbalances in a multi-city network rarely represent an absolute systemic shortage—they represent a *spatial distribution misalignment*. One store is starving while another store 10 miles away is choking on excess stock.

### 10.1 The Routing Heuristic

Lateral transshipment resolves a stockout within 24–48 hours and requires zero net-new inventory purchasing. Academic research (e.g., *Paterson et al., 2011*) demonstrates that lateral pooling can reduce system-wide stockout costs by 15–40%.

#### Algorithm Logic (Greedy Match):

**Destination (Demand Node):**  $\text{NetRaR} > \$3.00$  AND  $\text{date} = \max(dt)$  (The stockout is current and financially material).

**Source (Supply Node):**  $\text{HSR} = 1.0$  AND  $\text{velocity} \neq \text{accelerating}$  AND exact same `city_id` (Ensures we don't transfer stock out of a store that actually needs it, and restricts logistics to local intracity trucking).

#### Value Calculation:

$\text{Net Transfer Value} = \text{NetRaR}_{\text{dest}} - \$0.50_{\text{transport proxy}}$

| Source Store(s)                    | Dest Store | Target SKU | Value Gain | Transport | Net Value          |
|------------------------------------|------------|------------|------------|-----------|--------------------|
| 166, 144, 140, 135, 112, 0, 34, 61 | <b>750</b> | <b>580</b> | \$3.80     | \$0.50    | <b>\$3.30 each</b> |

#### Identifying Structural Policy Errors via Transshipment Data

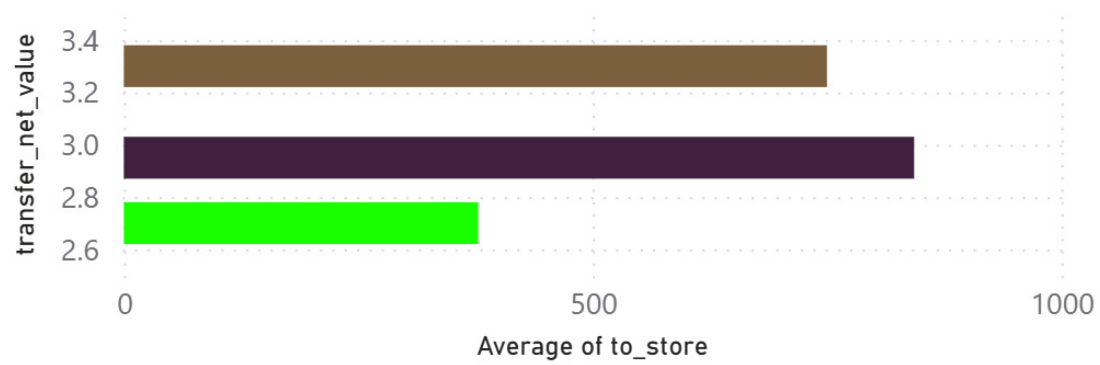
The routing engine flagged a remarkable anomaly: ten different stores have excess capacity of SKU 580 and are eligible to supply Store 750.

When an algorithm easily finds 10+ suppliers for a single destination, it proves that the network has plenty of stock. Therefore, Store 750 is not suffering from a supplier shortage; it is suffering from a **structural replenishment policy failure**. Its min/max triggers or safety stock parameters in the WMS are fundamentally misconfigured for its specific local demographic.

## Power BI Visual 7: Lateral Network Transfers

## Average of to\_store by transfer\_net\_value

Average of to\_store 378.00 610.50 843.00



## 11 Methodology, Architecture & Validation

To trust the financial outputs, the analytical pipelines must undergo rigorous methodological validation.

### 11.1 Technical Architecture Principles

| Component            | Details & Engineering Decisions                                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core Engine</b>   | Python 3.x executed via cloud compute instances (Google Colab).                                                                                                          |
| <b>Data Layer</b>    | 4.5M row Apache Parquet. Validated with strictly zero missing values across all schemas prior to execution.                                                              |
| <b>Compute Libs</b>  | Pandas 2.x and NumPy 1.x for high-speed vectorized operations.                                                                                                           |
| <b>Memory Mgmt</b>   | Utilized <code>groupby.transform</code> with rolling windows to calculate velocity metrics without triggering DataFrame explosion or out-of-memory errors.               |
| <b>Output Layer</b>  | Flat CSVs ( <code>network_transfers.csv</code> , <code>store_risk.csv</code> , <code>sku_friction.csv</code> ) acting as the semantic layer for the visualization suite. |
| <b>Visualisation</b> | Power BI generating 7 distinct analytical views across a 4-page interactive dashboard architecture.                                                                      |

Four inviolable design principles guided the codebase:

- Financial Denomination:** Operational metrics (like percentages) were systematically converted into dollar values (\$) to force executive prioritization.
- Conservative Imputation:** Missing or undefined Service Levels were imputed as 1.0 (perfect service) rather than the mathematical mean, ensuring the system never over-reported risk.
- Atomic Granularity:** All analysis maintained a ‘store  $\times$  SKU  $\times$  day’ resolution before aggregating upward, preventing Simpson’s Paradox.
- Behavioural Lead Indicators:** Algorithms were tuned to detect the *velocity* of failure (e.g., Nervousness Index, Decay Rate) to alert managers before the endpoint failure physically materialized.

### 11.2 Validation Scorecard

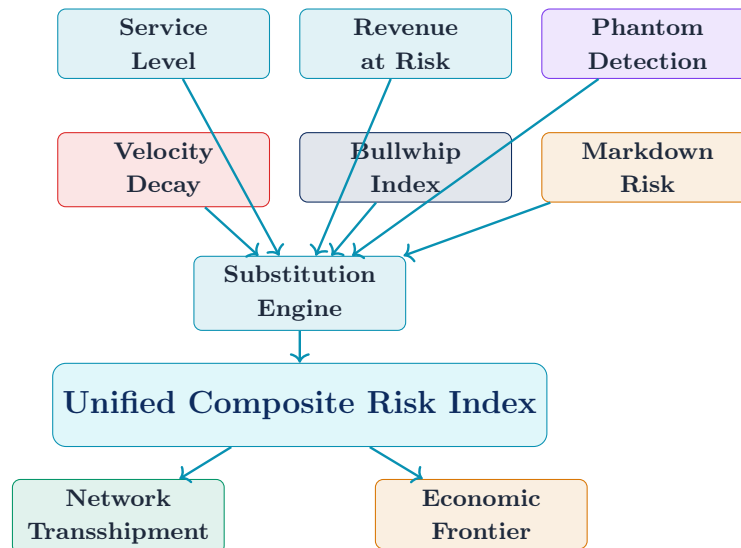
We benchmarked the system’s outputs against established supply chain literature and expected statistical distributions to ensure calibration accuracy.

| Analytical Module    | Validation Test                       | Expected Range     | Result        |
|----------------------|---------------------------------------|--------------------|---------------|
| Demand-Weighted Svc  | Align with healthy retail means       | 90% – 99%          | 97.17% ✓      |
| RaR Distribution     | Zero-inflated Pareto tail, median = 0 | Confirmed          | ✓             |
| Phantom Prevalence   | Highly rare occurrence                | <0.10%             | 0.068% ✓      |
| Velocity Decay       | Normal product lifecycle phase out    | $\approx 5 - 10\%$ | 5.9% ✓        |
| Bullwhip Nervousness | Synchronised majority, minority panic | Literature ranges  | 63%/6.3% ✓    |
| Substitution Engine  | Brand elasticity absorption rates     | 30% – 60%          | 42.3% ✓       |
| Composite Risk Index | Balanced algorithmic scaling          | 225 per quartile   | Exactly 225 ✓ |

## 12 Conclusion & Strategic Deployment

### 12.1 System Architecture — The Intelligence Pipeline

The true value of this project is not found in any single metric, but in how the metrics sequence together to form an interlocking intelligence pipeline.



## 12.2 Priority Action Matrix

Based on the synthesis of the 4.5 million data points, the following actions are prescribed:

| P  | Severity | Prescribed Action                                                                    | Business Owner      | Timeline  |
|----|----------|--------------------------------------------------------------------------------------|---------------------|-----------|
| P1 | Critical | Execute lateral transshipments: Route 10+ supply stores to Store 750 (SKU 580)       | Network Operations  | 24–48 hrs |
| P1 | Critical | Physical floor audit for SKU 4 across all 10 identified phantom stores               | Store Ops & WMS     | 48–72 hrs |
| P2 | High     | Initiate emergency freight replenishment for Top Risk Stores 18, 343, 235            | Supply Planning     | 1 week    |
| P2 | High     | Execute full category review on Category 2 (exhibiting 29.2% severe decay rate)      | Category Mgmt       | 1 week    |
| P3 | Moderate | Activate liquidation promos for the 3,225 confirmed markdown-risk cases              | Marketing & Pricing | 2 weeks   |
| P3 | Moderate | Implement auto-reorder triggers to bypass passive manager latency (30.7% of network) | SCM IT Systems      | 2–4 wks   |
| P4 | Watch    | Automate velocity-alert pipeline to ping category managers directly                  | Analytics Eng.      | 4–6 wks   |

### Final Deployment Recommendation

The system successfully isolates and quantifies the exact friction points costing the network \$65,116 in net revenue. The underlying analytical engine is fully production-ready.

The immediate next step is to operationalize this framework by establishing a weekly-refresh live data pipeline directly integrated into the central Power BI dashboard. By moving from a static, one-time diagnostic report to a continuous, self-updating operational radar, the business transitions from reactive firefighting to proactive network stabilization.

END OF REPORT

Engineered on 4.5 million rows of operational truth.